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Palaeomagnetic results from late Archaean and early Proterozoic granites from South Western Kenya

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Abstract - Results of a palaeomagnetic study carried out on the Migori granites exposed in the South Western part of Kenya are presented. The granites of Post-Nyanzian and Post- Kavirondian ages are dated at 2739 ± 111 Ma and 2420 ± 60 Ma, respectively, using the whole rock Rb-Sr isochron method. Alternating frequency and thermal demagnetization techniques have been used to isolate characteristic components of natural remanence. Opaque petrological and thermomagnetic studies indicate titanomagnetite grains as the main carrier of the primary component. The mean palaeomagnetic poles for the Post-Nyanzian and the Post-Kavirondian granites are located at 336°E, 72°N (δm = 12°, δP = 6°) and 263°E, 41°N (δm = 23°, δP = 15°), respectively.

INTRODUCTION

The first Proterozoic palaeomagnetic results from the East African craton were published by Patel and Raja (1979). A palaeomagnetic pole for the Mumias granites, dated at 2530 ± 50 Ma (Rb-Sr), was computed at 210.7°E, 60.9°N, (A_{95} = 13.5°). McElhinny *et al.* (1968) initially proposed the African Polar Wander path for the interval 2700-2350 Ma using palaeomagnetic data from the Zimbabwe and Kaapvaal cratons. A single apparent Polar Wander swathe for the interval 2300 - 1950 Ma was published by Piper et *al.* (1973) using the data from the Congo, the West African and the Kaapvaal cratons. Recently, Layer (1986) reported a palaeomagnetic pole for the (Archaean) Usushwana Complex (2870 ± 40 Ma) from Swaziland, at 347°E, 9°N, (A_{95} = 7.6°). The other six published African palaeomagnetic poles of late Archaean to early Proterozoic age are derived from the Modipe gabbro (2630 ± 470 Ma, Botswana), the Great dyke (2520 ± 30 Ma, Zimbabwe), the Jeppestown lava (2500 Ma, South Africa), the Gabarones granites (2340 ± 50 Ma, Botswana), the Lower Ventersdorp lavas (2500 Ma, South Africa) and the Upper Ventersdorp lavas (2300 ± 100 Ma, South Africa) as quoted by Piper (1976).

All palaeomagnetic data in the interval of 2900-2300 Ma are from Southern Africa except that of the Mumias granites from Kenya. The aim of the present study of the Migori granites from the South Western Kenya is to extend the palaeomagnetic studies of rocks back to late Archaean times and hence to add more information for defining the polar wander path over the interval 2800-2300 Ma.

GEOLOGY

The oldest rocks in the Migori area belong to the Nyanzian and the Kavirondian systems. The Kavirondian system unconformably overlies the Nyanzian system, which is believed to include the oldest rock in Kenya. These two systems are intruded by granites. According to Huddleston (1951) all the granite masses in this area only postdate the Nyanzian system. However, recent Rb-Sr dates on the granites indicate that they are of Post-Nyanzian and Post-Kavirondian age (Yanagi and Suva, 1981).

The Post-Nyanzian granites, the Mohoru-Lolgorian mass (Fig. 1.), have been dated at 2739 ± 111 Ma by Cahen *et al.* (1984) and at 2800 ± 120 Ma by Dodson *et al.* (1975). Both ages are derived from the whole-rock Rb-Sr isochron method. The Post-Kavirondian granites, (the Oyugis, Wanjare, Kitere and Kilgoris masses) are believed to be of Post-Kavirondian age. The Wanjare mass has been dated at 2420 ± 60 Ma (Yanagi and Suva, 1981). In hand specimen the Kitere granites are fine grained, pale or pinkish grey and moderately porphyritic with pink or greenish feldspar. The Wanjare granites are highly leucocratic, usually coarse grain and often porphyritic, pink or pale grey. The Kilgoris granites show no significant variations from the Wanjare mass. The Mohuru-Lolgorian mass is non-porphyritic and displays monzonitic texture. They are white and greenish in hand

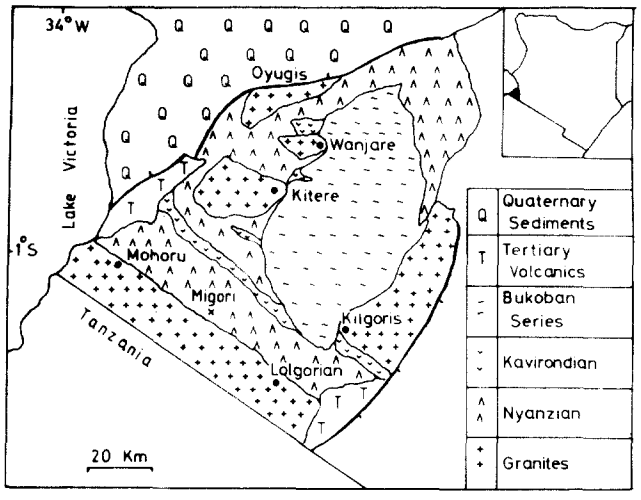


Fig. 1. Simplified geological map illustrating the occurrences of the granites masses and the locations of sampling sites.

specimens. Green hornblende is the principal dark mineral and is usually associated with greenish-brown biotite (Shackleton, 1946).

SAMPLING AND MEASUREMENTS

Freshly exposed granitic rock in this region is hard to find owing to lack of road-cuttings. Most of the outcrops were found to be typically boulderly and unsuitable for palaeomagnetic studies. A total of twenty four oriented block samples was collected from the four masses of granite. Samples for sites A and B were collected from the Mohoru and Lolgorian areas, respectively, whereas, samples for sites C, D and E were collected from the Wanjare, Kitere and Kilgoris areas as shown in Fig. 1. Sample collection at each site was spread out over distances of the order of a kilometer along the roads. The sample locations were usually from bridge-construction sites where the granite was reasonably well exposed.

Four to five core samples of 2.2 cm diameter were drilled from each of the oriented block samples and specimens of 2.5 cm length were cut from the cores. Their natural remanant magnetizations (NRMs) were measured on a Schonstedt digital spinner magnetometer. Three to four specimens from each sample were subjected to progressive alternating frequency (af) demagnetization and one specimen from each sample was thermally demagnetized. The Curie temperature analyses were carried out in air using a conventional Curie balance with a magnetizing field of 636.8 x 10^3 A/M. Polished samples were also prepared from each sample for petrological study.

RESULTS AND DISCUSSIONS

Characteristic af demagnetization curves are shown in Fig. 2. Mean destructive fields (mdfs)

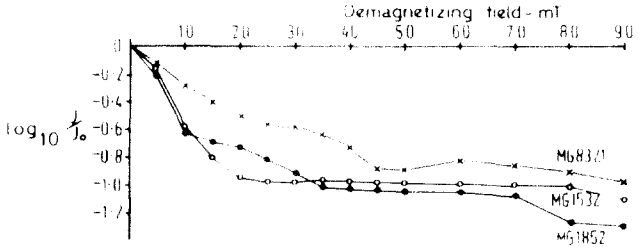


Fig. 2. Typical normalized alternate frequency demagnetization curves.

range from 10-15 mT. Typical orthogonal thermal demagnetization plots (z-plots) for specimen possessing stable and unstable components are illustrated in Fig. 3. The orthogonal plots were computed using methods described by Schmidt (1982). A few specimens were discarded which behaved erratically and gave no end-directions. A typical representative orthogonal plot of such a specimen is illustrated in Fig. 3c. This type of behaviour is not uncommon in Proterozoic rocks (Patel and Raja, 1979). Stability index criteria was used in deriving the end points as described by Briden (1972).

In the analysis samples were given unit weight. Each sample mean represent average of at least four end-directions. The combined af and thermal vectors of each sample block were treated for randomness at 5 % significance point. Any sample which failed this test was rejected from the analysis. The combining of the af and the thermal data for the sample mean may be justified by two ways. First, there are stable directions upon the thermal demagnetization as illustrated by Fig. 3a and 3b. Second, the thermal and the af end-directions agreed with each other at 5 % significance point. The paleomagnetic results of the Post-Kavirondian and the Post-Nyanzian granites are tabulated in Table 1.

Typical thermomagnetic curves are shown in Fig. 4. The Curie temperatures varied from 500°C to 550°C. The polished section petrology showed that the specimens of all of the sites

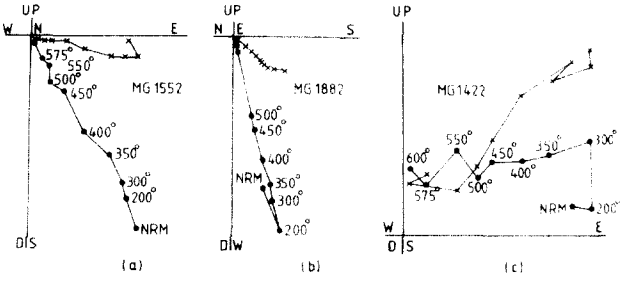


Fig. 3. Characteristics orthogonal demagnetization plots from specimens of the granites. Closed circles, vertical plane, crosses, horizontal plane. E-east, W-west, N-north, D-down

Table 1. Site mean directions and palaeomagnetic poles of the granite masses from South Western

	After a.f. & Thermal demagnetization					Paleomagnetic North Pole			
	N	Mean D.	Mean I.	R	k	α95	Log.(E)	Lat.(N)	δm	δp
Post-Nyanzian Granites										
Site A	5	333.5	+15.9	4.92	50.1	10.9	323.4	62.2	11.2	5.8
Site B	4	359.3	+17.0	3.87	23.9	19.2	30.3	80.6	19.8	10.2
Combined (A & B)	9	344.8	+16.8	8.59	19.7	11.9	336.2	72.3	12.3	6.3
Post-Kavirondian Granites										
Site C	4	342.4	-45.7	3.92	36.2	15.4	245.8	58.5	19.8	12.6
Site D	5	291.6	-29.4	4.86	30.1	14.2	288.0	20.8	15.6	8.6
Site E	4	332.6	-57.2	3.82	16.7	23.1	245.7	45.1	33.6	24.5
Combined (C, D & E)	13	319.2	-49.1	11.1	6.3	17.8	263.4	41.3	23.7	15.7

N = Number of unit vectors
D = Declination, East of T.N., degrees
I = Inclination, Positive down, degrees
R = Resultant Vector
k = Fisher's precision parameter
α95 = Radius of 95 per cent circle of confidence, degrees
Log., Lat. = Longitude & Latitude, degrees
δm, δp = Semi axes of 95% oval of confidence, degrees

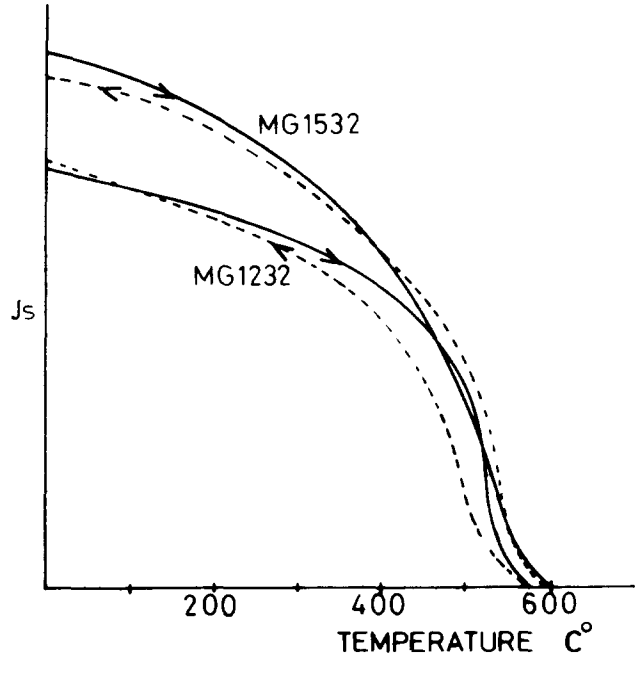


Fig. 4. Representative thermomagnetic curves (J_s Vs T). J_s is the saturation magnetization in arbitrary units. Continuous lines are heating curves; dotted lines are cooling curves.

contained titanomagnetite grains about 150-200 μm in diameter. The specimens from the fine grained variety of the granites had moderate degrees of exsolution by ilmenite lamellae. The thermomagnetic curves also confirm that magnetite is the main carrier of the natural remanence in the granites. The depression in the Curie temperature of magnetite maybe due to contamination by titanium.

The overall mean direction for the Post-Nyanzian granites is calculated at D = 344.8°, I = +16.8° (α_{95} = 11.9°) using nine unit vectors, whereas, for the Post-Kavirondian it is calculated at D = 319.2°, I = -49.1° (α_{95}=17.8°). The mean directions are unlike the present Earth's field in the Migori area (Skinner, 1975). The mean direction of the Post-Kavirondian granites (2420 ± 60 Ma) is not significantly different at the 5 % level from the direction of the Mumias granites (Patel and Raja, 1979) which is dated at 2530 ± 50 Ma. The SouthWestern part of Kenya escaped the influence of the Pan-African thermal event found in the Mozambique Belt in Kenya during the interval 450-700 Ma (Cahen *et al.*, 1984). Consequently, the cleaned directions of the Post-Kavirondian and the Post-Nyanzian granites of the Migori area maybe regarded as the ancient geomagnetic directions predating the Pan-African events.

The mean palaeomagnetic pole for the Post-Nyanzian granites is calculated at 336.2°E, 72.3°N (δm = 12.3°) δp = 6.3°), and the pole for the Post-Kavirondian granites is located at 263.4°E, 41.3°N (δm = 23.7°, δp = 15.7°).

CONCLUSION

The validity of the Mumias granites pole, the first Proterozoic pole from the East African craton, was discussed by Patel and Raja (1979). The poles obtained for the Post-Kavirondian and Post-Nyanzian granites from the present study, are located at 263°E, 41°N (δm = 23°, δp = 15°) and at 336°E, 72°N (δm = 12°, δp = 6°), respectively. Unfortunately all the granite masses of Post-Kavirondian age are not dated. The large error in the pole of Post-Kavirondian age, may be attributed to time variations in intrusion of these granite masses as observed by Pulfrey (1945). The age

determinations by Cahen *et al.* (1984) and Dodson *et al.* (1975) of the Post-Nyanzian granites are similar.

According to Cahen *et al.* (1984), cratons developed in Africa at different times in different places between around 3400 Ma and 2450 Ma. Any conclusions on a possible relative motions between the East African craton and the Transvaal or Zimbabwe cratons might not be detected at the present time because there is an insufficient number of palaeomagnetic poles of Archaean to early Proterozoic age. Secondly, most of the poles published have large errors in excess of ten degrees. Consequently, small relative motions might also not be detected (Briden, 1976).

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